TOYSOUND-01

Low-Power Digital Sound Controller ASIC

Project 01 — COB ASIC Laboratory

Technology:	SCL Open Source 1.2 μm
Design type:	Digital ASIC core
Application:	Toy / Sound Module / Low-Cost Consumer Electronics
Version:	v0.1
Date:	October 2, 2026

```
ASIC DIE
   |
WIRE BOND
   |
COB PCB
   |
SPEAKER
   |
TOY PRODUCT
```

Contents

1 Project Status Dashboard **3**

2 Document Control **3**

3 Executive Summary **3**

4 Product Idea **4**

5 Requirements **4**

6 System Block Diagram **4**

7 Circuit Diagrams **4**
- 7.1 A. Button interface . . . 4
- 7.2 B. Debounce concept . . . 5
- 7.3 C. Clock divider . . . 5
- 7.4 D. Output stage . . . 5

8 Digital Logic Diagram **5**

9 FSM Diagram **5**

10 RTL File Documentation & Programming Logic **6**
- 10.1 toysound_pkg.sv . . . 6
- 10.2 toysound_top.sv . . . 6
- 10.3 button_debounce.sv . . . 6
- 10.4 sound_controller.sv . . . 6
- 10.5 sound_sequencer.sv . . . 6
- 10.6 timer.sv . . . 7
- 10.7 tone_generator.sv . . . 7

11 RTL Source Tree **7**

12 Testbench Documentation **8**

13 Simulation File Types **8**
- 13.1 .sv . . . 8
- 13.2 .v . . . 8
- 13.3 .sv testbench . . . 8
- 13.4 .vcd . . . 8
- 13.5 .vvp . . . 8
- 13.6 .log . . . 8
- 13.7 .fst . . . 8
- 13.8 waveform images . . . 8

14 Waveform Section **8**

15 Simulation Flow Diagram **9**

16 FPGA Section **9**

17 ASIC Section **9**

18 PDK Section **9**

19 Library Section **9**

20 Synthesis Report **10**

21 Gate-Level Netlist **10**

22 Floorplan **10**

23 Placement **10**

24 Clock Tree Synthesis **10**

25 Routing **10**

26 STA **10**

27 DRC **10**

28 LVS **10**

29 Final GDSII **10**

30 Tapeout **10**

31 Fabrication **11**

32 Die **11**

33 Die Testing **11**

34 COB Packaging **11**

35 COB Cross-Section **11**

36 Pin / Pad Specification **11**

37 Datasheet **11**

38 Application Circuit **12**

39 PCB Design **12**

40 PCB BOM **12**

41 Final Product Functionality **12**

42 Test Procedure **12**

43 Failure Analysis **12**

44 Cost Model **12**

45 Design File Index **12**

46 Artifact Relationship Diagram **13**

1 Project Status Dashboard

Phase	Status
SPECIFICATION	[COMPLETE]
ARCHITECTURE	[COMPLETE]
RTL	[COMPLETE]
SIMULATION	[COMPLETE]
FPGA	[PENDING]
ASIC SYNTHESIS	[PENDING]
PHYSICAL DESIGN	[PENDING]
DRC	[PENDING]
LVS	[PENDING]
GDSII	[PENDING]
TAPEOUT	[PENDING]
FABRICATION	[PENDING]
DIE TEST	[PENDING]
COB	[PENDING]
PCB	[PENDING]
FINAL PRODUCT	[PENDING]

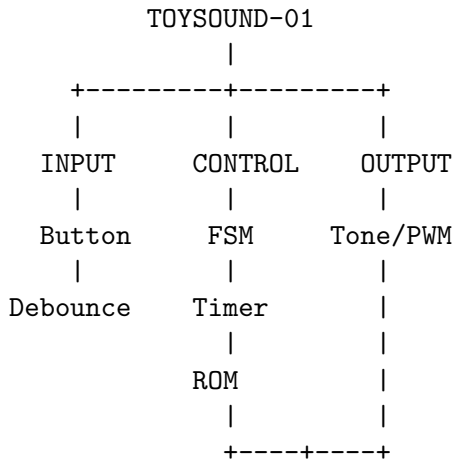
2 Document Control

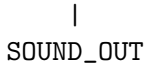
Version	Date	Change	Status
v0.1	October 2, 2026	Initial architecture	Draft
v0.2		RTL completed	
v0.3		Simulation verified	
v0.4		FPGA verified	
v0.5		ASIC synthesis	
v0.6		Physical design	
v0.7		DRC/LVS	
v1.0		Tapeout	
v2.0		Silicon result	

3 Executive Summary

Explain TOYSOUND-01 in one or two pages.

Example architecture:





4 Product Idea

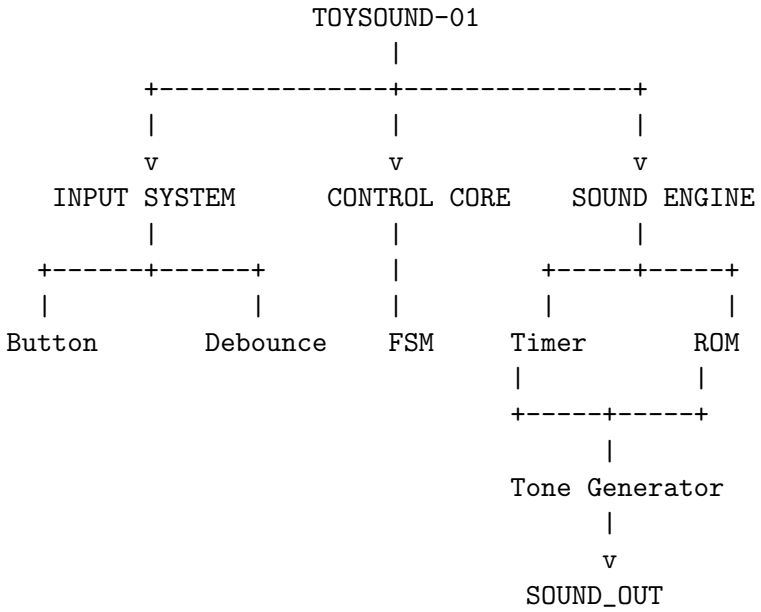
Explain the original product concept. Potential applications: toy sound module, educational toy, talking toy, button-operated sound toy, simple alarm, doorbell, electronic book, low-cost consumer sound controller.

Current TOYSOUND-01: 4-button simple tone playback. **Future product variants:** PCM audio, melodies, interactive games.

5 Requirements

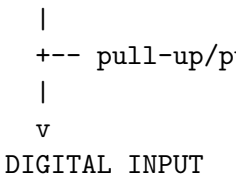
ID	Requirement	Target	Status
R01	Digital controller	Required	COMPLETE
R02	Button inputs	1	COMPLETE
R03	Sound output	1	COMPLETE
R04	Reset	Required	COMPLETE
R05	Low power	Target	TBD
R06	SCL 1.2 μm ASIC	Target	PLANNED
R07	FPGA prototype	Required	PLANNED
R08	DRC	Required before tapeout	PLANNED
R09	LVS	Required before tapeout	PLANNED
R10	COB packaging	Future	PLANNED

6 System Block Diagram



7 Circuit Diagrams

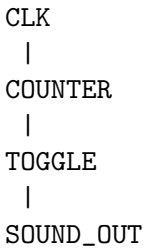
7.1 A. Button interface



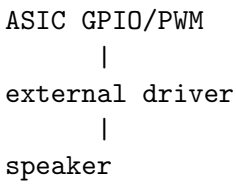
7.2 B. Debounce concept

Show the digital debounce logic: A sequential counter filtering unstable high-low states.

7.3 C. Clock divider

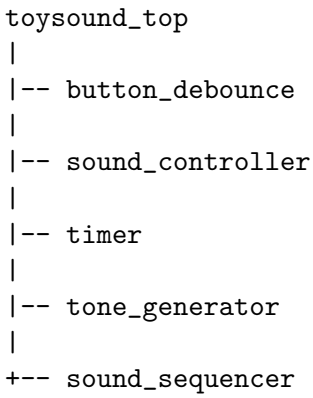


7.4 D. Output stage

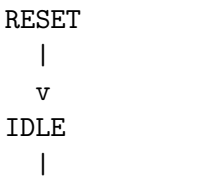


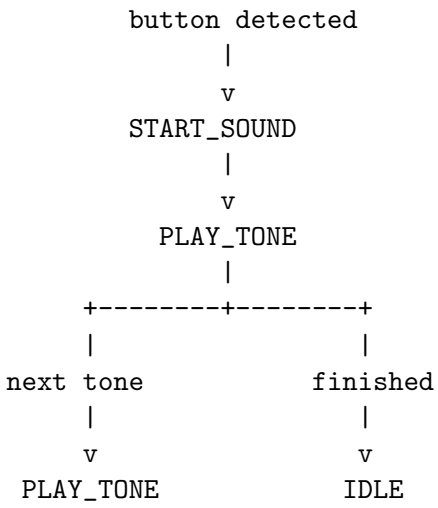
Note: The ASIC does not directly drive the speaker. An external BJT/MOSFET driver is required.

8 Digital Logic Diagram



9 FSM Diagram





10 RTL File Documentation & Programming Logic

The SystemVerilog code is partitioned carefully to isolate physical timing from logical state machines. Here is the programming essence of each file:

10.1 toysound_pkg.sv

Purpose: Global definitions.
Programming: Contains `parameter` constants defining musical note frequencies (as integer dividers of the 1 MHz clock) and durations. It also defines the `tone_t` struct, ensuring all modules agree on data widths.

10.2 toysound_top.sv

Purpose: Top-level hardware wrapper.
Programming: Contains no behavioural logic. It simply instantiates the lower-level modules and wires them together. It exposes only the 6 physical pins that will be bonded to the PCB.

10.3 button_debounce.sv

Purpose: Filter mechanical button bounce.
Programming: Uses an internal counter. When the button is pressed, the counter increments. If the button bounces (voltage fluctuates), the counter resets. It only emits a `btn_event` pulse once the signal is stable for a programmed number of clock cycles.

10.4 sound_controller.sv

Purpose: Main Finite State Machine (FSM).
Programming: Implements a 4-state FSM (`IDLE`, `LOAD_SOUND`, `PLAYING`, `DONE`). It maintains a `song_index` counter that increments on every button press. It commands the sequencer to start and then ignores the button until the song finishes.

10.5 sound_sequencer.sv

Purpose: ROM lookup and sequence timing.
Programming: Instead of using expensive SRAM memory, the songs are programmed as a giant `case` statement. The synthesis tool optimizes this into a tiny cluster of combinational logic

gates. It steps through the notes, sending the required frequency divider to the tone generator and waiting for the timer to signal completion.

10.6 timer.sv

Purpose: Measures note durations.
Programming: A simple countdown register. Loaded with a 20-bit value representing milliseconds (e.g., 200,000 clock cycles = 200ms). When it hits zero, it fires a `done` pulse back to the sequencer.

10.7 tone_generator.sv

Purpose: Square wave output generation.
Programming: Receives a frequency divider number. It counts down from that number to zero, then toggles the `sound_out` pin state (HIGH → LOW). This continuous toggling creates the PWM square wave that drives the speaker.

File	Module	Purpose	Inputs	Outputs
toysound_top.sv	toysound_top	Top integration	clk, reset_n, button	sound_out, busy
button_debounce.sv	button_debounce	Filter mechanical bounce	clk, reset_n, btn_raw	btn_event
sound_controller.sv	sound_controller	Main FSM	clk, reset_n, btn_event	sound_sel, play_en
timer.sv	timer	Countdown timer	clk, reset_n, load	done
tone_generator.sv	tone_generator	Square wave output	clk, reset_n, divider	tone_out
sound_sequencer.sv	sound_sequencer	ROM reader	clk, reset_n, start	tone_div, tone_en

11 RTL Source Tree

```
rtl/
|-- top/
|   `-- toysound_top.sv
|
|-- core/
|   |-- sound_controller.sv
|   |-- sound_sequencer.sv
|   `-- timer.sv
|
|-- peripherals/
|   |-- button_debounce.sv
|   `-- tone_generator.sv
|
```

```
‘-- pkg/
    ‘-- toysound_pkg.sv
```

12 Testbench Documentation

Test	Description	Expected	Actual	Status
TB01	Reset	Known state	PASS	COMPLETE
TB02	Button 1	Sound 1	PASS	COMPLETE
TB03	Button 2	Sound 2	PASS	COMPLETE
TB04	Button 3	Sound 3	PASS	COMPLETE
TB05	Button 4	Sound 4	PASS	COMPLETE
TB06	Bounce	Clean event	PASS	COMPLETE
TB07	Reset during sound	Safe recovery	PASS	COMPLETE

13 Simulation File Types

13.1 .sv

SystemVerilog source.

13.2 .v

Verilog source.

13.3 .sv testbench

Verification environment.

13.4 .vcd

Value Change Dump. Contains signal transitions over simulation time.

13.5 .vvp

Compiled simulation representation executed by vvp.

13.6 .log

Simulation/tool logs.

13.7 .fst

Waveform database format.

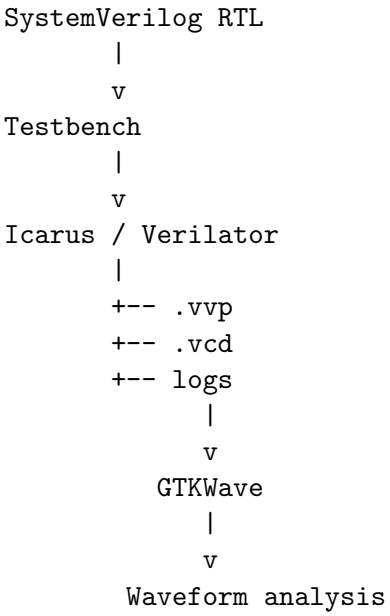
13.8 waveform images

PNG/PDF screenshots generated from GTKWave.

14 Waveform Section

STATUS: NOT YET IMPLEMENTED

15 Simulation Flow Diagram



16 FPGA Section

STATUS: NOT YET IMPLEMENTED

`RTL -> FPGA synthesis -> Implementation -> Bitstream -> FPGA -> Buttons + speaker`

The SCL ASIC PDK is not used for FPGA implementation.

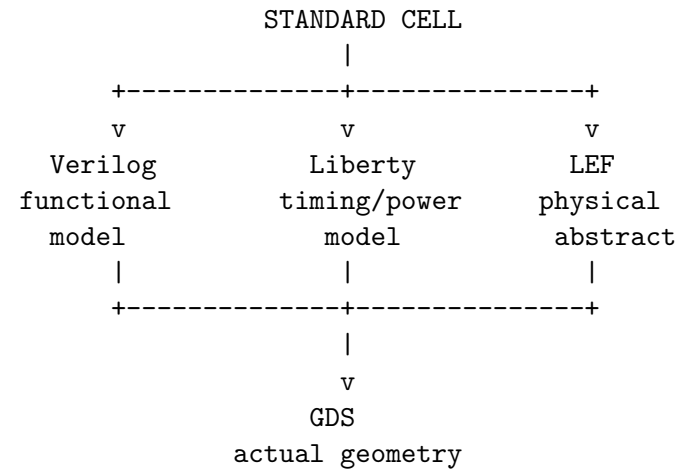
17 ASIC Section

STATUS: NOT YET IMPLEMENTED

18 PDK Section

Technology: SCL 1.2 μm PDK (C1D). Contains Liberty, LEF, GDS, Verilog cell models, DRC/LVS rule decks, and RCX extraction.

19 Library Section



20 Synthesis Report

STATUS: NOT YET IMPLEMENTED

21 Gate-Level Netlist

STATUS: NOT YET IMPLEMENTED

22 Floorplan

STATUS: NOT YET IMPLEMENTED

23 Placement

STATUS: NOT YET IMPLEMENTED

24 Clock Tree Synthesis

STATUS: NOT YET IMPLEMENTED

25 Routing

STATUS: NOT YET IMPLEMENTED

26 STA

STATUS: NOT YET IMPLEMENTED

27 DRC

STATUS: NOT YET IMPLEMENTED

28 LVS

STATUS: NOT YET IMPLEMENTED

29 Final GDSII

STATUS: NOT YET IMPLEMENTED

30 Tapeout

STATUS: NOT YET IMPLEMENTED [ ] RTL frozen [ ] Verification complete [ ] Synthesis complete [ ] Timing signoff [ ] Physical implementation [ ] DRC clean [ ] LVS clean [ ] GDS generated

31 Fabrication

STATUS: AWAITING FABRICATION

32 Die

STATUS: AWAITING FABRICATION

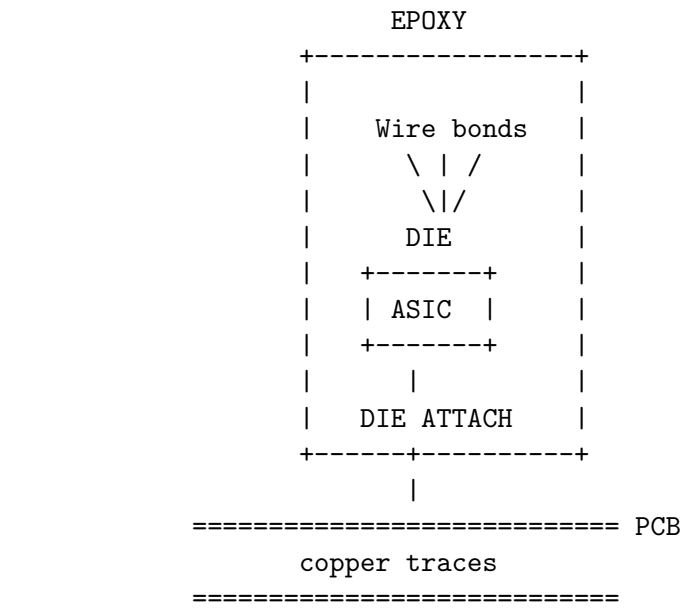
33 Die Testing

STATUS: NOT YET IMPLEMENTED

34 COB Packaging

```
Bare Die -> Die attach -> Wire bonding -> Electrical connection to PCB -> Encapsulation -> E
```

35 COB Cross-Section



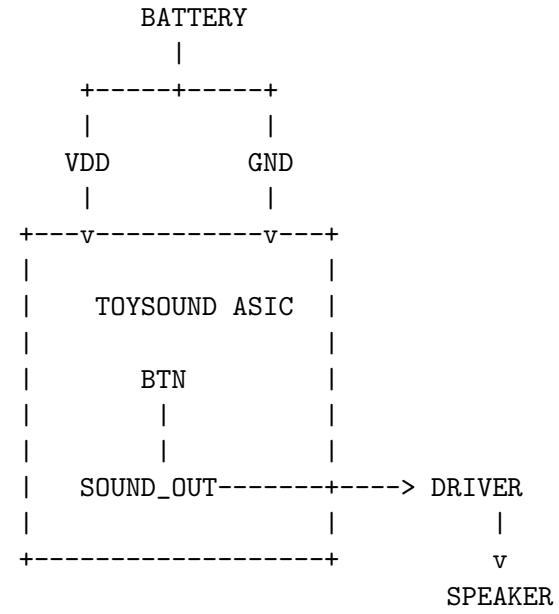
36 Pin / Pad Specification

Pin	Name	Direction	Function	Electrical Notes
1	VDD	Power	Supply	TBD
2	GND	Power	Ground	TBD
3	CLK	Input	Clock	TBD
4	RESET	Input	Reset	TBD
5	BTN	Input	Button (Next)	TBD
6	SOUND_OUT	Output	Sound signal	TBD

37 Datasheet

STATUS: TBD

38 Application Circuit



39 PCB Design

STATUS: NOT YET IMPLEMENTED

40 PCB BOM

STATUS: NOT YET IMPLEMENTED

41 Final Product Functionality

STATUS: NOT YET IMPLEMENTED

42 Test Procedure

STATUS: NOT YET IMPLEMENTED

43 Failure Analysis

STATUS: NOT YET IMPLEMENTED

44 Cost Model

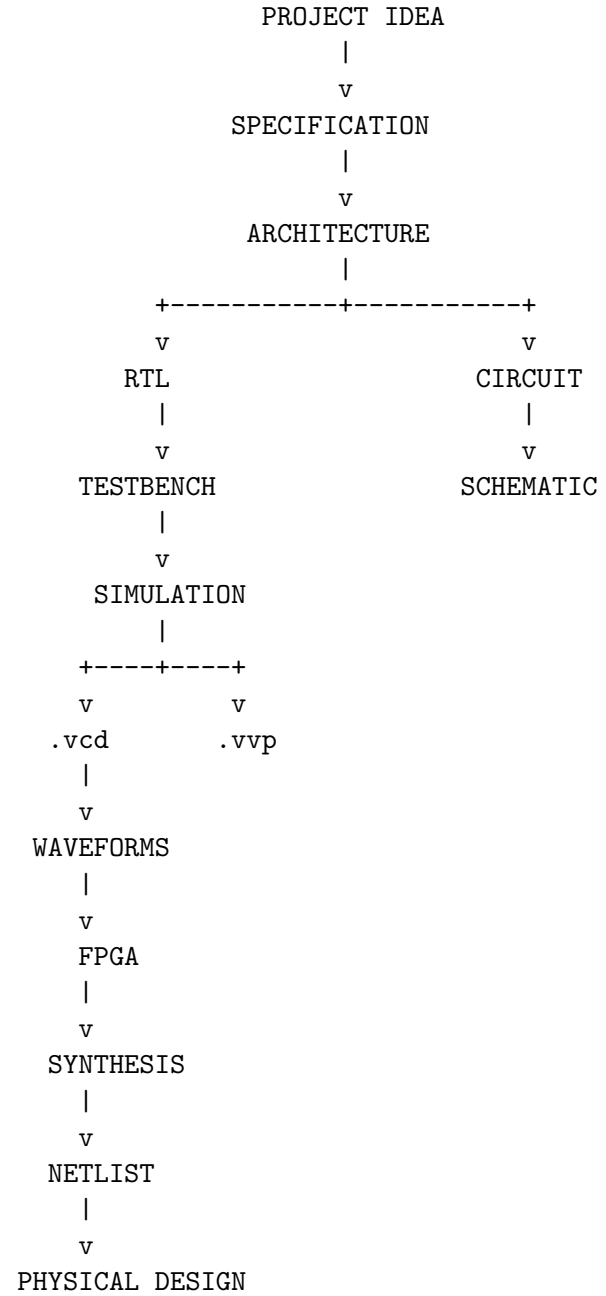
STATUS: NOT YET IMPLEMENTED

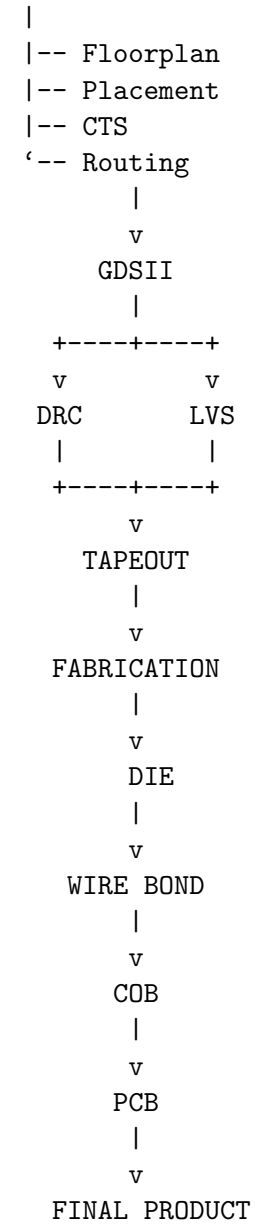
45 Design File Index

File	Type	Purpose	Status
rtl/core/sound_controller.sv	RTL	Logic module	Complete

File	Type	Purpose	Status
rtl/core/sound_sequencer.sv	RTL	Logic module	Complete
rtl/core/timer.sv	RTL	Logic module	Complete
rtl/core/tone_generator.sv	RTL	Logic module	Complete
rtl/peripherals/button_debounce.sv	RTL	Logic module	Complete
rtl/pkg/toysound_pkg.sv	RTL	Logic module	Complete
rtl/top/toysound_top.sv	RTL	Logic module	Complete
tb/toysound_top_tb.sv	Testbench	Verification	Complete
sim/toysound.vvp	VVP	Simulation artifact	Generated
sim/results/toysound_top.vvp	VVP	Simulation artifact	Generated
sim/waveforms/toysound_top_tb.vcd	VCD	Simulation artifact	Generated

46 Artifact Relationship Diagram




|
|-- Floorplan
|-- Placement
|-- CTS
`-- Routing
GDSII
DRC
LVS
TAPEOUT
FABRICATION
DIE
WIRE BOND
COB
PCB
FINAL PRODUCT